(i)

$$F_0 = 0 \implies a + b = 0 \implies b = -a$$

$$F_1 = 1 \implies a(\lambda - \mu) = 1$$

$$F_2 = 1 \implies a(\lambda^2 - \mu^2) = 1 \implies \lambda + \mu = 1$$

$$F_3 = 2 \implies a(\lambda^3 - \mu^3) = 2$$

$$a(\lambda - \mu)(\lambda^2 + \lambda\mu + \mu^2) = 2$$

$$\lambda^2 + \lambda\mu + \mu^2 = 2$$

$$\lambda^2 + \lambda\mu + \mu^2 = (\lambda + \mu)^2 - \lambda\mu$$

$$\lambda\mu = -1$$

$$\begin{aligned}(t - \lambda)(t - \mu) &= t^2 - (\lambda + \mu)t + \lambda\mu \\ &= t^2 - t - 1\end{aligned}$$

So λ, μ are roots of

$$t^2 - t - 1 = 0 \implies t = \frac{1 \pm \sqrt{5}}{2}$$

$$\lambda = \frac{1 + \sqrt{5}}{2},\ \mu = \frac{1 - \sqrt{5}}{2},\ a = \frac{1}{\sqrt{5}},\ b = -\frac{1}{\sqrt{5}}$$

(ii)

$$F_6 = \frac{1}{\sqrt{5}}\left(\lambda^6 - \mu^6\right)$$

$$\lambda^2 = \lambda + 1,\ \mu^2 = \mu + 1 \implies \lambda^6 = 8\lambda + 5,\ \mu^6 = 8\mu + 5$$

$$F_6 = \frac{1}{\sqrt{5}}\left[(8\lambda + 5) - (8\mu + 5)\right] = \frac{8(\lambda - \mu)}{\sqrt{5}} = 8$$

(iii)

$$\begin{aligned}\sum_{n=0}^{\infty}\frac{F_n}{2^{n+1}} &= \frac{1}{2\sqrt{5}}\sum_{n=0}^{\infty}\left[\left(\frac{\lambda}{2}\right)^n-\left(\frac{\mu}{2}\right)^n\right]\\ &= \frac{1}{2\sqrt{5}}\left(\frac{1}{1-\frac{\lambda}{2}}-\frac{1}{1-\frac{\mu}{2}}\right)\\ &= \frac{1}{\sqrt{5}}\cdot\frac{\lambda-\mu}{(2-\lambda)(2-\mu)}\end{aligned}$$

$$\lambda-\mu=\sqrt{5}$$

$$(2-\lambda)(2-\mu)=4-2(\lambda+\mu)+\lambda\mu=1$$

$$\sum_{n=0}^{\infty}\frac{F_n}{2^{n+1}}=1$$

This solution was submitted by **Jon**
Published and refined by **Jon**

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